

Exploring generative AI in the misinformation Era: Impacts as a misinformation source and fact-checker on belief in the information

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ABSTRACT

This research aims to explore the impact of generative AI on public belief in misinformation both as a source and as a fact-checker, and to examine how individual factors such as AI literacy and trust in AI influence these perceptions applying machine heuristics. The study conducted two online experiments focusing on health and nutrition issues. The first experiment assessed whether generative AI as a source of misinformation influences public belief differently compared to traditional internet sources. The second experiment evaluated the effectiveness of generative AI in the role of a fact-checker compared to human fact-checkers. The results reveal no significant differences in the belief in misinformation when it was presented by generative AI versus other internet sources. However, human fact-checkers were found to be perceived as more trustworthy than their AI counterparts especially health and nutrition issues, confirming negative machine heuristic. Notably, the level of trust in AI significantly affected participants' acceptance of AI-generated fact-checking, particularly concerning claims about breastfeeding and child intelligence. This study contributes to the ongoing discourse on the role of AI in misinformation management, highlighting the complex interplay between technological innovations and human cognitive biases.

1. Introduction

The rise of online platforms and social media has provided numerous benefits to communication and information gathering; however, it also made it easier to spread misinformation, drawing significant scholarly attention (Chaudhuri et al., 2024; Gurgun et al., 2024). The widespread impact of misinformation became especially evident during COVID-19 pandemic (Gisondi et al., (2022)), and similar patterns have emerged across other issues, such as climate change (Lewandowsky, 2021), vaccination (Wang et al., 2019) and genetic modified food (Bode et al., 2021). In response, scholars and practitioners have sought ways to curb the spread of misinformation on social media. However, with the recent rise of generative AI, a new challenge has surfaced: generative AI is now a tool that can create, rather than just spread, highly convincing misinformation.

Generative AI, which produces content like text and images based on patterns from training data (Martineau, 2024), has become

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widely accessible through models like ChatGPT. This technology can generate highly credible misinformation, making it even harder to distinguish truth from falsehood (Xu et al., 2023). Unlike before, when misinformation primarily spread through sharing, we now face an era where people can actively generate convincing misinformation themselves. Experts are concerned that generative AI will exacerbate this problem, as it allows for the creation of more authentic-looking false information (Bail, 2024). Another caveat is that the public holds varying perceptions of AI—while some believe AI is unbiased and precise (Binns et al., 2018), others are growing wary of its potential to mislead, especially as concerns about AI's misuse in misinformation have been raised (Lee Rainie, 2022).

In this study, we aim to investigate the impact of generative AI on the perception of misinformation across various contexts. Our first objective is to determine whether misinformation produced by AI affects people's belief a concern widely shared by experts. Additionally, we examine how individual factors related to AI technology, such as AI literacy, influence these beliefs. Given that individuals with extensive knowledge about AI are likely to have a deeper understanding of its mechanisms (Wang et al., 2023; Wang et al., 2023), this could be a critical factor in evaluating AI-generated misinformation. By exploring the influence of AI as a source of misinformation and identifying individual factors that might affect this influence, we aim to provide insights on how to effectively address misinformation in the era of generative AI.

2. Literature review

2.1. Influence of generative AI on misinformation beliefs: Applying machine heuristics

The widespread adoption of internet technologies and the ubiquity of social media platforms have introduced a new era characterized by the frequent diffusion of misinformation (Kumar & Geethakumari, 2013). New media platforms enable individuals to generate and share content without gatekeeping, often leading to disseminate incorrect information (Vinhas & Bastos, 2022). This phenomenon has become a serious issue, as it misled public behavior in critical domains like public health (Gisondi et al., (2022)), climate change (Lewandowsky, 2021) and vaccination (Wang et al., 2019). Nowadays, individuals can commonly encounter misinformation while searching information online. However, this trend faces new challenges with the advent of generative AI.

Generative AI refers to machine learning models that can generate new content, ranging from text and images to music and code, based on the patterns and information it has learned from training data (Martineau, 2024). This capability is fundamentally different from traditional AI models that are primarily designed for analysis and prediction. The powerful capabilities of generative AI have gained significant attention due to their potential applications in various fields, from creative industries to automated customer service. However, this hype also brings challenges, particularly in the realm of misinformation.

Generative AI can play an important role in spreading misinformation. It can produce highly realistic fake texts, images, and videos, including deepfakes, which can dangerously spread false information and manipulate public opinion (Chesney & Citron, 2019; Zellers et al., 2019). The ability of generative AI to generate realistic content makes it difficult for individuals to distinguish between real and fake information (Kreps et al., 2022), thus exacerbating the spread of misinformation. In this era of misinformation, understanding how individuals respond to the misinformation that is generated by AI will help gauge the current stage of public's understanding and reactions and develop effective strategies to combat misinformation.

To understand the impact of generative AI on misinformation spread across the web, the concept of machine heuristics (Sundar & Kim, 2019; Yang & Sundar, 2024) can offer insights. According to the machine heuristics, there are two different types of machine heuristics: (1) positive heuristic and (2) negative heuristic. The positive heuristic is the belief that machines are more objective and error-free than humans. More specifically, the authority heuristic posits that machines and automated systems are often perceived as authoritative and trustworthy sources of information (Molina & Sundar, 2022). For instance, Binns et al. (2018) found that participants thought AI could be more precise and fairer than humans because it consistently applies the same rules. This belief can lead users to trust machines more, assuming that their judgments are objective and accurate (Sundar, 2020). Especially with the rapid advancement of AI, especially in areas like product recommendations, AI-labeled recommenders have gained trust comparable to human-labeled ones (Liu & Wang, 2024). Given these insights, individuals with a positive machine heuristic might be more likely to believe in even misinformation if they believe it was generated by AI.

However, there could be another possibility especially considering the negative machine heuristics. The negative heuristic is the belief that machines lack nuanced subjective judgment (Lee, 2018). Many perceive AI as inflexible and unable to understand complex cognitive mechanism of human, therefore, have a lower level of trust in its decision. For instance, research by (Hong et al., 2024) indicates a preference for human journalists over AI, suggesting a broader skepticism towards machine capabilities in roles traditionally filled by humans. Such findings suggest that in the realm of generative AI and misinformation, negative heuristics could lead to a backlash against AI-generated information, reducing its perceived credibility. In addition to these theoretical discussions, recent polls also show increasing scrutiny over AI's limitations and misinformation potential among the public. Pew Research (Lee Rainie, 2022) finds 45 % of U.S. adults equally concerned and excited about AI in daily life, citing potential job loss, privacy risks, and misuse. In addition, a Forbes survey (Haan, 2024) revealed over half of consumers think they can distinguish between human and AI-generated content, yet 76 % worry about misinformation from AI tools like Google Bard and ChatGPT.

In this study, we will compare how people respond to misinformation generated by AI through the lens of positive or negative machine heuristics by comparing their reactions to AI generated misinformation versus general web-based misinformation. As these heuristics typically compare machines (such as generative AI) to human beings, their applicability to web-based misinformation might initially seem questionable, since one could argue that all web content is, to some extent, machine-mediated. However, recent research demonstrated that people generally perceive web-based information as human-generated unless explicit cues indicating the source or creator of the content exist (Jakesch et al., 2023). This insight guides our comparison between general web-based misinformation and

misinformation explicitly labeled as AI-generated, as people are likely to attribute the source of unlabeled information to humans by default. As labeling AI-generated content has been commonly discussed as a strategy to reduce the risks of generative AI (Wittenberg et al., 2024), we are interested in how explicit AI labels stand out among general online information without arbitrary human labels, impacting user perceptions. We speculate when such cues are present, machine heuristics may come into play, leading people to view the information as a product of systematic, non-human processes.

To sum up, given the ongoing debate and recent findings, there are two potential responses to AI-generated misinformation. When individuals encounter potential misinformation from AI, they may apply either positive or negative machine heuristic. On one hand, the positive machine heuristic might dominate, driven by belief in AI's objectivity and unbiasedness. This could lead people to trust even misinformation if it is generated by AI. On the other hand, growing skepticism towards generative AI, particularly concerning its role in spreading misinformation, might lead people to adopt a negative machine heuristic. In this case, skepticism could prevail, causing people to distrust and scrutinize the information more critically. Considering these contrasting perspectives, we raise the following research question: if particularly labeling misinformation as AI generated will significantly change people's beliefs. Particularly, in practice, the vast majority of non-AI-generated information remains unlabeled, making it difficult to establish a clear contrast between "human" and "AI" sources. Therefore, rather than comparing AI-labeled content to explicitly human-labeled information, we examine whether AI-labeled content stands out against the average, unlabeled information found online.

RQ1: Are people more likely to believe in misinformation generated by AI compared to misinformation without a clearly labeled sources?

2.2. The moderating role of AI literacy

The concept of AI literacy is crucial in understanding how individuals respond to misinformation generated by AI. Although AI literacy is a relatively new concept, the broader concept of literacy in media and communication technologies is well-established, extending beyond traditional reading and writing to include competencies related to navigating and critically evaluating media content (Buckingham & Burn, 2007; Gilster and Glistler, 1997). This expansion of literacy now involves the ability to access, analyze, and evaluate messages from various media platforms (Aufderheide, 1993; Silverblatt, 2001).

Research indicates that literacy levels significantly impact how individuals interpret media content. For example, those with higher media literacy are better at identifying fake news and display greater skepticism towards its credibility (Lu et al., 2022). Similarly, individuals with advanced news media literacy are more critical of news shared on social media (Vraga & Tully, 2021). These findings suggest that literacy in specific media technologies influences how information is assessed and interpreted. A similar mechanism is likely to be applied to AI.

AI literacy, according to Wang et al., 2023; Wang et al., 2023, includes not only the capacity to understand and use AI technology but also the ability to apply these technologies effectively for data analysis, selection, and evaluation, while maintaining an ethical perspective. This literacy encompasses subdimensions such as awareness, usage, evaluation, and ethics (Long & Magerko, 2020; Ng et al., 2021; Wang et al., 2023; Wang et al., 2023). Among these, AI evaluation is particularly significant in how misinformation generated by AI is assessed. Unlike basic AI awareness and usage, which involve surface-level understanding and practical applications, AI evaluation entails a critical engagement with AI-generated information, assessing its capabilities and limitations (Ng et al., 2021; Wang et al., 2023; Wang et al., 2023). Hence, individuals proficient in AI evaluation could be equipped to scrutinize the content and context of information produced by AI more thoroughly, enhancing their ability to judge its validity. This deeper understanding of AI's operations and its broader implications is essential when confronting AI-generated misinformation, influencing cognitive processes significantly more. Therefore, we propose:

H1. The effect of AI-generated misinformation (vs other sources) on belief in such misinformation will be dependent upon an individual's AI awareness literacy level; specifically, those with higher AI literacy are less likely to believe misinformation from AI compared to those with lower literacy levels.

3. Method

3.1. Procedure

To answer the research questions and test the proposed hypothesis, we designed a between-subjects experiment focusing on misinformation sources: AI-generated misinformation versus general internet information (control). We manipulated the information source by labeling it as either generated by AI or derived from a general internet information center. To select relevant topics for this study, we referenced subjects previously tested to determine if human beings or AI-based natural language models respond differently to potentially deceptive questions (Lin et al., 2021). From 817 items, we initially identified topics that were adversarial, incorrect, and contained clearly opposing "truths" which was crucial step in determining the clear misinformation to be used in our experiment. Among those topics, we particularly focused on health and nutrition topics because these are areas where individuals increasingly seek information online (Jia et al., 2021; Ruani et al., 2023), yet also frequently encounter misinformation (Denniss et al., 2023; Schmid et al., 2023). Given that these topics are both highly sought after and susceptible to misinformation, they align well with our study's objective to explore the public's discernment of accurate versus misleading content. After an initial review by two researchers, a total of 12 issues were selected for inclusion in the study (see Appendix A for a list of the 12 issues).

To refine the selection of topics for our study, we conducted a pre-test with 298 participants from the Prolific panel. The goal of this pre-test was to identify which of the 12 topics participants perceived as misinformation while also feeling uncertain about their accuracy. This step was crucial because the primary goal of our study is to determine whether the label “Generative AI” influences perceptions of misinformation. Uniformity in perceiving a specific topic as misinformation is important – if some participants view a topic as misinformation while others see it as accurate, other factors, such as their prior beliefs, rather than the source cue, may drive the evaluation, making it difficult for us to conclude whether the belief in the misinformation is due to the source or other factors.

However, at the same time, we wanted the topics to contain a degree of ambiguity. Specifically, we selected topics that were generally perceived as misinformation but not with absolute certainty – claims that seemed questionable and left room for doubt about exactly why or how they were inaccurate. This uncertainty would encourage participants to rely on cues, such as the source of the information, to evaluate its credibility. In other words, rather than choosing claims that are universally and unquestionably dismissed as false, we deliberately focused on ambiguous or controversial misinformation statements – claims that appear dubious but still plausible. These gray-area topics are more susceptible to source effects because they prompt individuals to depend more on cues like whether the information originated from AI or another source when making judgements. Thus, selecting issues that were broadly regarded as misinformation yet contained elements of uncertainty would allow us to more effectively assess the impact of labeling a source as “generative AI”.

Based on a series of descriptive analyses (detailed in the [Appendix B](#)), we identified three key issues as the most suitable for testing the proposed hypotheses and addressing the research question: 1) the impact of breastfeeding on a child’s intelligence, 2) the determination of a baby’s sex during pregnancy, and 3) the use of coughing as a method to stop a heart attack. These issues were identified based on the findings that around 70 % to 90 % of participants perceived them as misinformation. Additionally, approximately half of the participants were certain about their assessment of the information, while the remaining half expressed uncertainty regarding their evaluation. For each of these issues, we created two versions of the information: one version cited artificial intelligence as the source, while the other indicated a general internet information source as the origin (control condition) (see [Fig. 1](#)).

Upon entering the online experiment, participants first evaluated their level of AI literacy. Subsequently, they were presented with information related to three issues selected above: the impact of breastfeeding on child intelligence, the determination of a baby’s sex during pregnancy, and using coughing as a method to stop a heart attack. The presentation order of these issues and the source type for each issue (either generative AI or a general internet source) were randomized. For example, a participant might first encounter information on the impact of breastfeeding attributed to generative AI, followed by information on coughing to stop a heart attack attributed to a general internet source, and finally, information on determining a baby’s sex also attributed to generative AI. Various combinations of issue order and source attribution were possible, but all participants were exposed to all three issues, each paired with a randomly assigned source type. Participants were then asked to assess how much they believe in the presented information. After evaluating all three issues, participants concluded the experiment by providing demographic information.

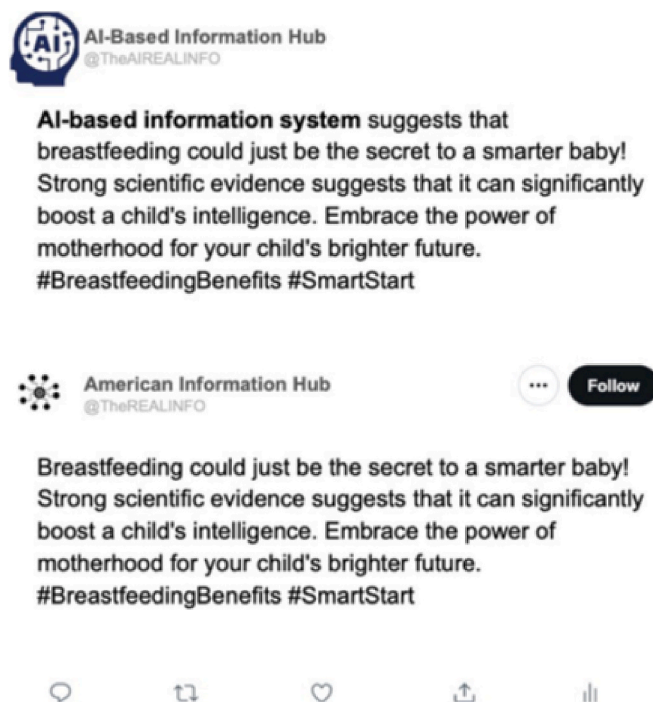


Fig. 1. Experimental stimuli used for the impact of breastfeeding on child intelligence issue.

3.2. Participants

The participants of this study were recruited via Prolific, totaling 200 individuals ($Mage = 40.41$, $SDage = 14.69$). Among them, 38.7 % ($n = 77$) were male, 59.8 % ($n = 119$) were female, and 1.5 % ($n = 3$) identified as others. Also, majority of the participants were White (65.8 %, $n = 131$), followed by Black (20.1 %, $n = 40$), Asian (6.5 %, $n = 13$), Hispanic (4.0 %, $n = 8$), and others (3.5 %, $n = 6$). However, participants who failed the manipulation check were excluded from the analysis. Specifically, individuals were excluded if they incorrectly identified the source of the information they were presented with—for example, stating the source was AI-based when they had seen information from a general online source, or vice versa. After these exclusions, the remaining participants counts for analysis were: 175 for the issue of the impact of breastfeeding on a child's intelligence (96 in the AI condition, 79 in the general internet source), 179 for the issue of determining a baby's sex during pregnancy (94 in the AI condition, 85 in the general internet source), and 177 for the issue of using coughing to stop a heart attack (100 in the AI condition, 77 in the general internet source). Only responses from participants who correctly answered the manipulation check question were included in further analyses.

3.3. Measures

3.3.1. Belief in misinformation

Assessed using five items modified from Flanagin & Metzger (2003). Sample items included, “the information that I read from the previous page is believable,” “The information that I read from the previous page is trustworthy.” All the responses were averaged to create an index of belief in misinformation (for breastfeeding issue $M = 4.01$, $SD = 1.65$, $\alpha = 0.93$, for determination of baby's sex issue, $M = 3.25$, $SD = 1.60$, $\alpha = 0.95$, and for heart attack issue $M = 3.72$, $SD = 1.58$, $\alpha = 0.94$).

3.3.2. AI literacy

Assessed with the three items adopted from Wang et al., 2023; Wang et al., 2023. Sample items include “I can evaluate the capabilities and limitations of an AI application or product after using it for a while”, “I can choose a proper solution from various solutions provided by smart agents”, and “I can choose the most appropriate AI application or product from a variety for a particular task.” We averaged all the responses to create an index of AI literacy (for breastfeeding issue $M = 4.98$, $SD = 1.21$, $\alpha = 0.82$, for determination of baby's sex issue, $M = 4.92$, $SD = 1.22$, $\alpha = 0.82$, and for heart attack issue $M = 4.96$, $SD = 1.21$, $\alpha = 0.81$).

3.4. Analyses

To address RQ1, we conducted an independent t -test to examine the differences induced by the source of the information. Subsequently, to explore the moderating role of AI literacy on the influence of generative AI as the source of misinformation on its belief, we utilized SPSS process Model 1 (Hayes, 2023). In this analysis, the experimental condition was dummy coded (1 = AI as the information source, 0 = control condition, which is the general internet information source).

4. Results

4.1. The impact of source of belief in misinformation

RQ1 sought to determine whether the source of information influences belief in misinformation. For the information related to breastfeeding, we found no significant difference in belief toward misinformation between AI ($M = 3.99$, $SD = 1.70$) and a general information source ($M = 4.17$, $SD = 1.60$), $t(173) = -0.71$, $p = 0.477$. Similarly, for the determination of a baby's sex issue, no significant difference was observed between AI as a source of misinformation ($M = 3.11$, $SD = 1.56$) and a general information source ($M = 3.41$, $SD = 1.63$), $t(177) = -1.28$, $p = 0.201$. Likewise, for the heart attack-related issue, no significant difference was found between AI as the source of misinformation ($M = 3.56$, $SD = 1.54$) and a general information source ($M = 3.94$, $SD = 1.63$), $t(175) = -1.61$, $p = 0.110$. Overall, our findings indicate that there were no significant differences in belief in misinformation, regardless of whether the source was AI or a general internet source, across all three topics.

4.2. The moderating role of AI-literacy

We posited in H1 that an individual's level of AI literacy would significantly moderate the impact of information source on belief in misinformation. However, the results indicated no significant moderating effect of AI literacy for the breastfeeding issue ($B = -0.26$, $SE = 0.20$, $p = 0.21$). Similarly for the determination of baby's sex issue ($B = -0.01$, $SE = 0.20$, $p = 0.38$) and heart attack issue ($B = -0.30$, $SE = 0.19$, $p = 0.12$), the interaction term of experimental condition and AI literacy was not significantly associated with credibility toward the misinformation, leading to the rejection of H1 for all issues.

5. Discussion

The goal of this study was to explore whether misinformation explicitly marked as AI generation would influence people's belief in such misinformation. We conducted an online experiment across three different topics to investigate this potential effect. Contrary to the perception that AI is unbiased and precise (Binns et al., 2018), we found no difference in the perceived credibility of AI-generated

misinformation when compared to general web misinformation. We also considered the possibility that, given the growing awareness of AI's role in generating misinformation (Lee Raine, 2022), participants might show a reduced belief in AI-generated misinformation. However, our findings did not support this hypothetical direction either; participants' belief levels were not significantly lower or higher for misinformation attributed to AI. These results suggest that AI, as a source label itself, does not influence people's perceptions of misinformation. Additionally, we found no significant effect of AI literacy on belief in misinformation.

Findings from our first research question offer clear implications and contribute meaningful value to the growing body of literature comparing AI-generated information with other sources. Given the rapid and widespread integration of generative AI into daily life, public perceptions and understanding of AI's role have shifted significantly over time even with a relatively short timeframe (Cheng et al., 2025). Exploring these perceptions in the very current context offers timely insight.

Our study reveals that misinformation labeled as "AI-generated" is not perceived as more or less credible than general web-based misinformation, suggesting that distinctions between AI and non-AI sources have significantly blurred. This may stem from the public's widespread familiarity with AI tools like ChatGPT, leading to the perception that most online content is at least partially AI-generated. Contrary to a couple of years ago, the public now often assumes that much online content is AI-generated. With around 90 % of Americans aware of AI (Faverio & Tyson, 2023) and AI increasingly embedded in socio-technical systems (Satori & Bocca, 2023), its distinctiveness as an information source has faded. This aligns with our findings- participants did not distinguish between AI-labeled and traditional web content in terms of credibility. Recent research further supports this, showing reduced differentiation between human- and AI-generated information (Huschens et al., 2023). Given that our data were collected in July 2024, during peak familiarity with platforms like ChatGPT, this shift likely reflects evolving public perception and marks a departure from earlier studies.

Also, from the perspective of both accumulation of related literature and theory building, these results challenge earlier studies that emphasized the strong influence of AI source cues on credibility perception (e.g., Liu and Wei, 2019; Wölker & Powell, 2021; Wu, 2022). This discrepancy may stem from our study's specific experimental design, which compared AI-generated misinformation to typical web-spread misinformation that lacked clear source attribution. While prior research demonstrates that users typically perceive unlabeled web content as human-generated (Jakesch et al., 2023) – potentially creating perceptual distinctions between such content and explicitly AI-labeled information – our findings challenge this established pattern. Our findings demonstrated that a clear "AI-generated" label and no-label condition did not make specific differences, challenging the concept of machine heuristics activation with the existence of a clear cue of "AI," aligning with recent studies' results regarding blurred boundaries between humans and machines (Liu & Wang, 2024).

In summary, our study offers a unique contribution by capturing a more contemporary view of AI's role in misinformation era. Rather than amplifying or diminishing credibility on its own, AI now blends seamlessly into the broader information ecosystem – an important shift that future misinformation research and public communication strategies must account for.

As research comparing AI and other types of content sources continues to grow, we believe it is important to deepen the discussion by examining individual-level factors related to AI, such as AI literacy. However, our findings revealed that AI literacy did not play a significant role. This lack of a significant moderating effect of AI literacy may also reflect the broader familiarity with AI technologies across individuals, regardless of their actual knowledge level. While those with high AI literacy are typically more informed and likely to assess information more systematically (Wang et al., 2023; Wang et al., 2023), our findings show minimal differences in how misinformation is perceived between more and less knowledgeable participants. One possible explanation is the Dunning-Kruger effect—where individuals with lower knowledge overestimate their competence (Dunning, 2011)—leading less informed participants to evaluate AI-generated content similarly to more knowledgeable ones. Another factor may be the use of self-reported AI literacy measures. Given the widespread exposure to terms like "generative AI" (Faverio & Tyson, 2023; Satori & Bocca, 2023), participants may have overestimated their actual literacy. To more precisely assess the role of AI literacy, future studies should incorporate objective or performance-based assessments that better capture true knowledge levels.

6. Study 2

In Study 1, we found that the AI source cue did not significantly influence misinformation assessment. In the subsequent study, we aim to determine if this effect persists when AI is used for fact-checking. While AI can generate damaging misinformation, it also has potential as an effective tool for countering misinformation when employed as a fact-checker (Kahn, 2024). This study differs from Study 1 in that we will focus on the distinction between "AI" and "human-based" fact-checkers, reflecting the core ideas of machine heuristics instead of comparing "AI" and other general web sources. In addition, this approach can ensure external validity as well, since it was observed that fact-checkers are often powered by AI LLMs and human experts (Liu et al., 2023). By examining AI's role in misinformation fact-checking, we seek to understand whether AI as a fact-checker shapes information assessment differently than humans, testing whether it triggers specific cognitive responses as suggested by machine heuristics or if the results mirror those of Study 1, where no significant effects were observed. This exploration could verify whether AI is perceived merely as an existing common technology with its unique impacts in misinformation diminishing, or if its influence varies in different roles and information contexts. Additionally, we will examine whether another individual factor related to AI, specifically AI trust, plays a role in the assessment of AI-generated fact-checkers in this particular study.

6.1. Effects of generative AI as fact-checkers: Applying machine heuristics

AI can serve both to generate realistic misinformation (Chesney & Citron, 2019; Zellers et al., 2019) and to combat it through automated fact-checking and content moderation, comparing information against verified sources to detect falsehoods and filtering

out misinformation on social media platforms (Baly et al., 2018; Graves, 2018; Hassan et al., 2017). Despite challenges like transparency and resource constraints, AI's role in fighting misinformation is increasingly recognized (Wolfe & Mitra, 2024).

Generative AI uses natural language processing (NLP), machine learning (ML), and data analytics in fact-checking to verify facts and generate explanatory content that makes interactions engaging and understandable. It excels in processing nuanced contexts and providing detailed responses (Baly et al., 2018). This AI continually improves by learning from vast datasets, potentially outperforming other fact-checkers in accuracy and relevance over time (Graves, 2018). By automating the identification of checkable statements and detection of previously verified information, AI significantly expedites the fact-checking process, allowing for real-time verification in multiple languages and formats, thus saving time and enhancing accuracy (Sohr & Piccato, 2024). Such efficiency supports positive machine heuristics in fact-checking contexts. For instance, Moon et al. (2023) found that participants showed less resistance to fact-checking messages from AI, even when they contradicted their political beliefs, reinforcing the perception of AI as an objective and impartial information source (Sundar & Kim, 2019; Sundar, 2020).

However, negative machine heuristics can also arise when using AI for fact-checking. As discussed in Study 1, AI-attributed messages can lower credibility in specific contexts (Shin, 2022; Xue et al., 2024). This scrutiny means the impact of certain factors like label intensity is more significant with AI-generated messages, as shown in a study by Tandoc et al. (2020), where AI presence did not necessarily enhance message credibility or trust. Considering these mixed findings, we would like to explore in this study, the effectiveness of generative AI as a fact-checker. Therefore, the following research question was suggested:

RQ2: Are people more likely to believe in fact-checker generated by AI compared to the fact-checker based on human experts?

6.2. The moderating roles of AI-trust

In Study 1, we investigated the role of AI literacy in assessing AI-generated misinformation. Despite theoretical relevance, AI literacy did not significantly moderate participants' responses to AI-generated content, possibly because self-reported measures poorly captured actual comprehension. Additionally, as most of users demonstrated moderately high AI literacy, AI literacy may not be an effective variable to differentiate users under the impacts of the source. AI literacy captures what users know about AI processes. However, knowing does not always mean endorsing. A user might understand large language models and still think they are dangerous, manipulative, or untrustworthy. As AI technologies become more ubiquitous and embedded in everyday platforms, general knowledge—regardless of depth—may overshadow formal distinctions in literacy. Therefore, we decided to focus on more predictive variables such as AI attitude rather than self-reported AI ability, which might have stronger and more consistent relationships with trust and user outcomes. In study 2, we replaced the moderator of interest from AI literacy to AI trust. AI trust offers a more direct assessment of how individuals perceive and rely on AI technologies. Trust is a crucial determinant in the adoption and use of technology and may provide more consistent insights into user behavior than literacy. This also better aligns with the aforementioned heuristic approach of perceiving AI: when evaluating AI-generated content, people often use heuristics ("Is this AI good or bad?") rather than systematically analyzing AI logic.

Trust in AI refers to the confidence that users place in artificial intelligence systems to perform tasks effectively, reliably, ethically, and in a manner aligned with human values. Trust in AI encompasses aspects like system transparency, accuracy, predictability, fairness, and the ability of the AI to operate safely in complex or uncertain environments (Mayer et al., 1995; Hoff & Bashir, 2015).

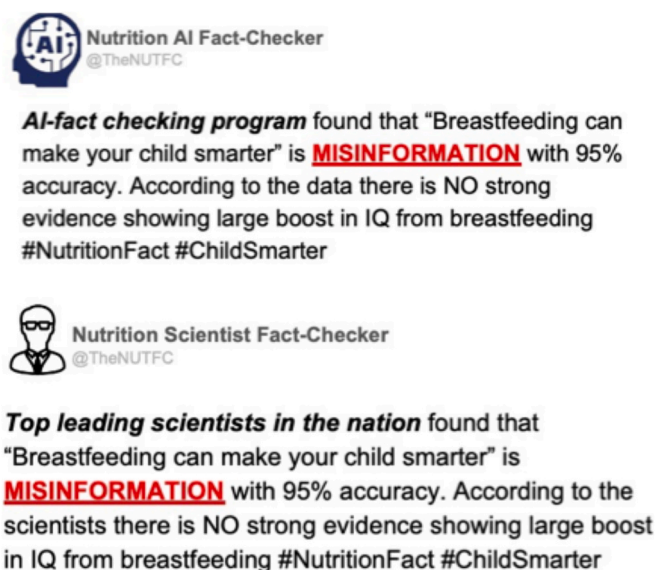


Fig. 2. Experimental stimuli used in Study 2 for the impact of breastfeeding on child intelligence issue.

According to Mayer et al. (1995), trust is “the willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor.” When applied to AI, this means that users trust AI when they believe it will act consistently and produce expected outcomes in a given context. Given the current mixed public sentiment toward AI, we propose that trust in AI is crucial when evaluating information fact-checked by either AI or human experts, as it may moderate the relationship between the type of fact-checker and perceived credibility. Based on this rationale, we postulate as below:

H2. The effect of AI-generated fact-checker (vs human experts) on belief in such fact-checker will be dependent upon an individual's AI trust level.

7. Study 2 method

7.1. Procedure

To address the research question and test the hypotheses proposed in Study 2, we conducted a between-subjects experiment focusing on the source of the fact-checker: AI-generated versus human expert fact-checkers. Study 2 utilized the same issues as Study 1, examining the misinformation related to the effects of breastfeeding on child intelligence, determining a baby's sex during pregnancy, and using coughing as a method to stop a heart attack. The social media posts in the study claimed that a specific fact-checker (either AI-generated or human expert) identified this information as misinformation (see Fig. 2).

In this experiment, participants first assessed their level of trust in AI. They were then presented with information on the three issues, which were randomly ordered. The source of the fact-checker for each issue was randomly assigned to participants, similar to Study 1. For example, a participant might first review information on the heart attack issue verified by an AI-generated fact-checker, followed by the breastfeeding impact evaluated by a human expert fact-checker, and finally, the determination of a baby's sex checked by an AI fact-checker. Various combinations of issue order and source attribution were possible, ensuring all participants were exposed to all three issues, each paired with a randomly assigned type of fact-checker. Participants were then asked to rate their belief in the fact-checker's ability. After evaluating all three issues, participants concluded the experiment by providing demographic information.

7.2. Participants

We recruited 200 participants from Prolific ($M_{age} = 35.15$, $SD_{age} = 11.40$). Among them, 36 % ($n = 72$) were male, 61 % ($n = 122$) were female, and 3 % indicated themselves as others ($n = 6$). Also 57.5 % ($n = 115$) were White, followed by 21 % Black ($n = 42$), 10 % Asian ($n = 10$), 2.5 % as others ($n = 5$). Among these participants, who failed the manipulation check were excluded from the analysis. For the manipulation check, we asked the participants whether the fact-checker was AI-generated fact-checker or human-expert based fact-checker. For those who indicated that the fact-checker was based on AI although they say human-expert based fact-checker, or the fact-checker was based on human-expert even though they saw AI-based fact-checker during the experiment were excluded from the further analysis. After these exclusions, 169 responses from the participants were left for breastfeeding issue (93 in the AI-fact checker 76 in the human-expert fact-checker), 175 were left for determining baby's sex during pregnancy (98 in the AI-fact checker 77 in the human-expert fact-checker), and 168 were left for heart attack issue (94 in the AI-fact checker 74 in the human-expert fact-checker). Only responses from participants who passed the manipulation check were used for the analyses to answer research questions and test hypotheses.

7.3. Measures

7.3.1. Belief in fact-checker

Assessed with the items adopted from Flanagin and Metzger (2003). The sample items include “The fact-checker that I saw above is accurate” and “The fact-checker that I saw above is trustworthy.” All the responses were averaged to create an index of belief in fact-checker (for breastfeeding issue $M = 4.41$, $SD = 1.73$, $\alpha = 0.95$, for determination of baby's sex issue, $M = 4.70$, $SD = 1.58$, $\alpha = 0.94$, and for heart attack issue $M = 4.86$, $SD = 1.94$, $\alpha = 0.94$).

7.3.2. AI trust

Assessed with four items from Liu (2021). Sample items include “If a decision is made by AI, it must be precise” and “If a decision is made by AI, it must be a good decision.” All the responses for the four items were averaged to create an index of AI trust, higher scores indicating higher level of trust in AI (for breastfeeding issue $M = 3.78$, $SD = 1.58$, $\alpha = 0.86$, for determination of baby's sex issue, $M = 3.82$, $SD = 1.64$, $\alpha = 0.87$, and for heart attack issue $M = 3.89$, $SD = 1.62$, $\alpha = 0.87$).

7.4. Analyses

To address RQ2, we conducted series of independent t-tests to examine the differences depending on the type of fact-checker (AI vs human experts). Subsequently, to explore the moderating roles of AI trust on the effect of generative AI as a fact-checker on its belief, we used SPSS Process Model 1 (Hayes, 2023). In this analysis, the experimental condition was dummy coded (1 = AI fact-checker, 0 = human-expert fact-checker).

8. Study 2 results

8.1. The effect of fact-checker type

RQ2 sought to determine which type of fact-checker would be perceived as more believable among the participants. For the information related to breastfeeding, we found that people have a stronger belief in fact-checker from human experts ($M = 4.94$, $SD = 1.35$) than AI based fact-checker ($M = 3.97$, $SD = 1.89$), $t(167) = -3.75$, $p < 0.001$. Similarly, for the determination of a baby's sex issue, people believed in the fact-checker based on human expert ($M = 5.09$, $SD = 1.48$) more than AI based fact-checker ($M = 4.29$, $SD = 1.59$); $t(173) = -3.01$, $p < 0.01$. Likewise, for the heart attack-related issue, human expert-based fact-checker ($M = 5.32$, $SD = 1.55$) was perceived as a more believable fact-checker than AI-based fact-checker ($M = 4.50$, $SD = 1.51$), $t(166) = -3.45$, $p < 0.001$. In sum, among all of the issues, participants in general believed that human-expert based fact-checker was more believable than AI-based fact-checker.

8.2. The moderating role of AI trust

H2, which suggested significant moderating role of AI trust, we found that AI-trust significantly moderated the impact of fact-checker type on belief in fact-checker ($B = 0.46$, $SE = 0.16$, $p < 0.01$) for the impact of breastfeeding on baby's intelligence issue. As can be seen from the below graph (see Fig. 3), participants who have a high level of trust in AI will be more likely to believe in the fact-checker by AI. However, for the issue of determination of baby's sex ($B = 0.18$, $SE = 0.14$, $p = 0.21$) and heart attack issue ($B = -0.08$, $SE = 0.15$, $p = 0.58$), AI trust did not play a significant role. Therefore, H2 was only supported for the breastfeeding issue.

9. Study 2 and overall discussion

In Study 2, generative AI as a fact-checker induced a difference in people's perceptions. We observed that participants placed more trust in human-based fact-checkers than those generated by AI. The discrepant results related to AI as a generator of misinformation from Study 1 and AI as a fact-checker from Study 2 can be explained by two main reasons. Firstly, while Study 1 compared AI and general information displayed on the web, Study 2 compared AI and humans as the source of fact-checkers. Therefore, machine heuristics could have fostered more strongly and induced differences between AI-generated and human-based fact-checker.

Furthermore, the nature of misinformation generation and misinformation could be different. The results from study 2 showed that humans seem to be more credible as information judges over AI. Such negative heuristics observed in the Study 2 seems to align with the argument made by Molina and Sundar (2024). They stated that humans are better suited for tasks requiring nuanced judgment, such as content moderation than machines. Additionally, Yang and Sundar (2024) suggest that the perceptions of AI's roles vary

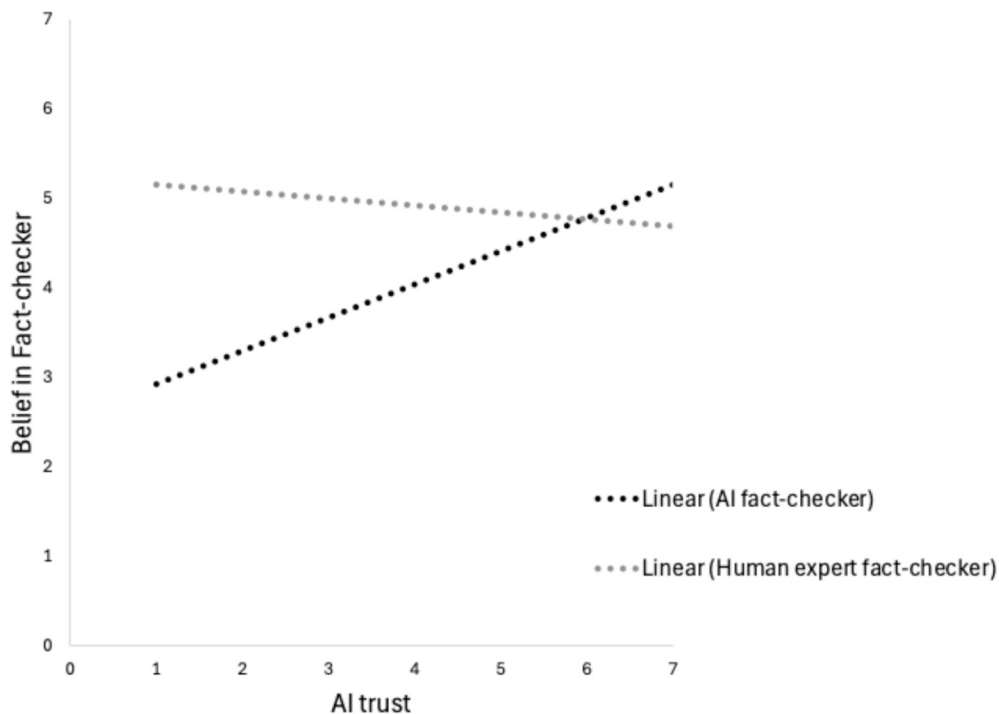


Fig. 3. Moderating Effect of AI Trust on the Influence of AI Fact-Checkers on Belief Regarding the Impact of Breastfeeding on Child Intelligence.

depending on whether the tasks are better suited for mechanical tasks (rule-based) or human tasks (subjective). Fact-checking, which involves the detailed assessment and synthesis of information, can be perceived as a complex task not well suited for AI's capabilities due to its requirement for nuanced understanding, not just simple rule application.

In addition, the focus on health and nutrition in this study might explain the pronounced negative heuristics observed. Lermann Henestroza and Kimmerle (2024) discovered that labeling content as AI-written diminishes its perceived credibility compared to human-authored content in the realm of science. A meta-analysis by Wang & Huang (2024) also highlighted that the subject matter significantly influences credibility perceptions, with machine-authored news on socio-political topics deemed less credible than human-written counterparts. This tendency suggests that negative machine heuristics are more likely to be triggered in contexts where the content is intimately connected to human concerns, as with the health-related topics in our study.

In addition, we observed that AI trust significantly moderated the impact of the AI fact-checker on belief, particularly concerning misinformation about breastfeeding's impact on baby's intelligence in Study 2. Comparing to other topics, this detected effect could have been due to the sample characteristics. The participants of the Study 2 were predominantly women (61 %), many of whom are of childbearing age (Chernikoff, 2024), as suggested by the mean age of our sample ($M_{age} = 35.15$). This demographic is likely to have a high level of involvement with information related to family and children issues. High involvement typically leads to more systematic processing of information, where individuals seek to understand the deeper meaning of the information presented (Lee & Kim, 2016; Petty & Cacioppo, 1986). Therefore, participants with high issue involvement who also trust AI may assume that the AI fact-checker has thoroughly accessed and accurately evaluated all relevant online information. This belief in the thoroughness and accuracy of AI fact-checkers makes them more likely to trust the AI-generated verdict on the misinformation. This suggests that the combination of personal relevance and trust in AI technology enhances their confidence in the fact-checker's credibility. More potential moderators about individual differences related to the specific information topics, such as gender, marital/parental status, and personal involvement, warrant future explorations.

To summarize, this study enriches our understanding of AI, particularly in the context of misinformation—a field fraught with both challenges and opportunities. It offers new insights on how AI can be effectively leveraged during this infodemic era. By empirically investigating the dual roles of generative AI as both a source and a checker of misinformation, the study aligns with and expands upon the theoretical framework of machine heuristics. It illustrates the multifaceted nature of AI—its varied forms, functions, and roles—and how these factors influence user trust dynamically. This highlights the ongoing challenge of accurately capturing the interactions between users and AI, particularly concerning the depth of information processing and the significance of the AI label in specific application contexts.

This research provides empirical evidence that, at the initial stage, when users take the content at face value, the presence of an AI label makes no significant difference compared to content without such a label. This may suggest that people are increasingly viewing AI as a parallel, though not necessarily distinct, information source from human-generated content. However, when AI is used for fact-checking or as an information arbiter, human judgment and preference still hold value. These findings offer practical insights for interface design and communication strategies, particularly on how labeling can be leveraged to build user trust effectively.

10. Limitations and future research

Despite offering the above insights, our study has limitations. In Study 1, unlike Study 2, we compared the role of AI as a source of misinformation against general internet-based information. We found significant distinctions between AI-generated fact-checker and human expert-based fact-checker in Study 2, leading us to hypothesize similar distinctions might have existed between AI-generated and human-generated misinformation. Given the widespread familiarity with generative AI (Faverio & Tyson, 2023; Satori & Bocca, 2023), the general public may no longer see AI as uniquely different from typical internet sources. Notably, search engines often prioritize AI-summarized information, potentially diminishing perceived differences between AI and general internet content. Yet, as indicated in Study 2, the interaction between AI-generated and human-generated misinformation warrants further exploration.

Additionally, AI trust may have influenced perceptions of AI-generated versus internet-generated misinformation, a factor unexplored in Study 1. Study 2 revealed that AI trust significantly influences perceptions when interacting with AI-based fact-checkers, particularly among individuals deeply engaged with the issue at hand. A similar impact may also be observable in the context of AI-generated misinformation. Therefore, future research should investigate how trust in AI affects the assessment of AI-generated misinformation.

Furthermore, our study utilized machine heuristics to suggest that there are two possible outcomes when engaging with generative AI in the realm of misinformation. To more accurately uncover the psychological mechanisms involved in responding to misinformation and fact-checking driven by generative AI, future research would benefit from directly measure individual's machine heuristics. This approach would allow for a more detailed exploration of how these heuristics influence perceptions and evaluations of misinformation and fact-checking processes.

Finally, in the experiments, we utilized health and nutrition-related issues to ensure external validity, as these are common topics for which individuals often seek information online (Jia et al., 2021; Ruani et al., 2023). We selected specific topics within the experiments, because they are widely recognized as misinformation, yet there remains uncertainty in public assessments, allowing us to examine the influence of source credibility on belief formation more clearly. However, given that the majority of prevalent misinformation and disinformation revolve around partisan topics, such as climate change or abortion (Lee & Seo, 2023), it is important that future research also explores these divisive issues to fully understand the dynamics of misinformation in different contexts. Meanwhile, incorporating factors such as personal relevance and issue involvement can help account for context dependency, offering a more nuanced understanding of these processes.

Furthermore, several factors may have influenced participants' assessments of the health and nutrition-related information, such as their level of involvement in health matters or trust in the medical system (Intemann, 2023; Tsai et al., 2018). For example, individuals with high involvement in health issues or strong trust in scientists may have been more inclined to rely on fact-checkers from human experts, as seen in the context of Study 2. Given these potential influences, we encourage future research to examine these variables to develop a more comprehensive understanding of the role of generative AI in the misinformation era.

Despite its limitations, this study adds important insights to the research on generative AI and public assessment. We found that generative AI isn't seen as a unique information category and doesn't differ notably from other internet sources. In line with recent meta-analyses (Huang & Wang, 2023), our findings show that AI doesn't significantly alter specific perceptions or attitudes more than other forms. This might change if AI's role shifts from generating to evaluating information. Results from our studies indicate that the public now discerns AI-generated information more keenly and perceptions vary with AI's role. As generative AI is increasingly used across social disciplines, deeper exploration of its roles is crucial for understanding its impact on public perceptions.

**During the preparation of this work the authors used ChatGPT-4o in order to improve readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.*

Author Note.

All authors have approved the manuscript and agreed with its submission to Telematics and Informatics. The authors are willing to share data, analytics methods, and study materials with other researchers. The material will be available upon request.

CRedit authorship contribution statement

Seo Yoon Lee: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Weizi Liu:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.tele.2025.102308>.

Data availability

Data will be made available on request.

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